Stochastic Recovery from Information Loss and Equality of Channel Capacity

Kirtus Justus

Theoretical Physicist

Port Coquitlam, British Columbia, Canada

Baseline edition, 12 November 2026

Contents

Part	Title	Page
1	Stochastic Recovery from Information Loss and Equality of Channel Capacity	2

Formal statements

Identifiers are those of the program theorem registry. Page numbers refer to this file.

Identifier	Statement	Page
F2-01-01	Theorem 2.1 (Information loss controls recovery)	3
F2-01-02	Theorem 3.1 (One full-support probe suffices for exact recovery)	4
F2-01-03	Corollary 4.1 (A full-support coarse-channel optimizer)	5
F2-01-04	Lemma 5.1 (A preserved convex relation obstructs recovery)	6
F2-01-05	Theorem 5.2 (Exact minimax error and anchor tradeoff)	6
F2-01-06	Proposition 6	8

Acknowledgments and contributions

Kirtus Justus directed the research questions, the selection of mathematical models and the publication standards, is the sole named author, and is responsible for the claims presented here. ChatGPT (OpenAI) assisted with derivations, critical checks and manuscript preparation. Claude (Anthropic), Gemini (Google), Grok (SpaceXAI) and DeepSeek (High-Flyer) contributed to the wider research process; attribution of particular arguments is limited to traceable records. Editorial preparation of this edition: Claude (Anthropic). Assistance does not imply provider endorsement, external peer review or institutional endorsement.